

## MILESTONE 1: CLIENT DESIGN REVIEW - CMPG325-2026-063

Date: 28 August 2026

Client: Pheny Cultural Tours (Potchefstroom)

Student: 46898064

### 1. CLIENT REQUIREMENTS

Organisation Type: **Tourism - Cultural Tours**, Bookings, Guided Experiences Business Needs

Identified:

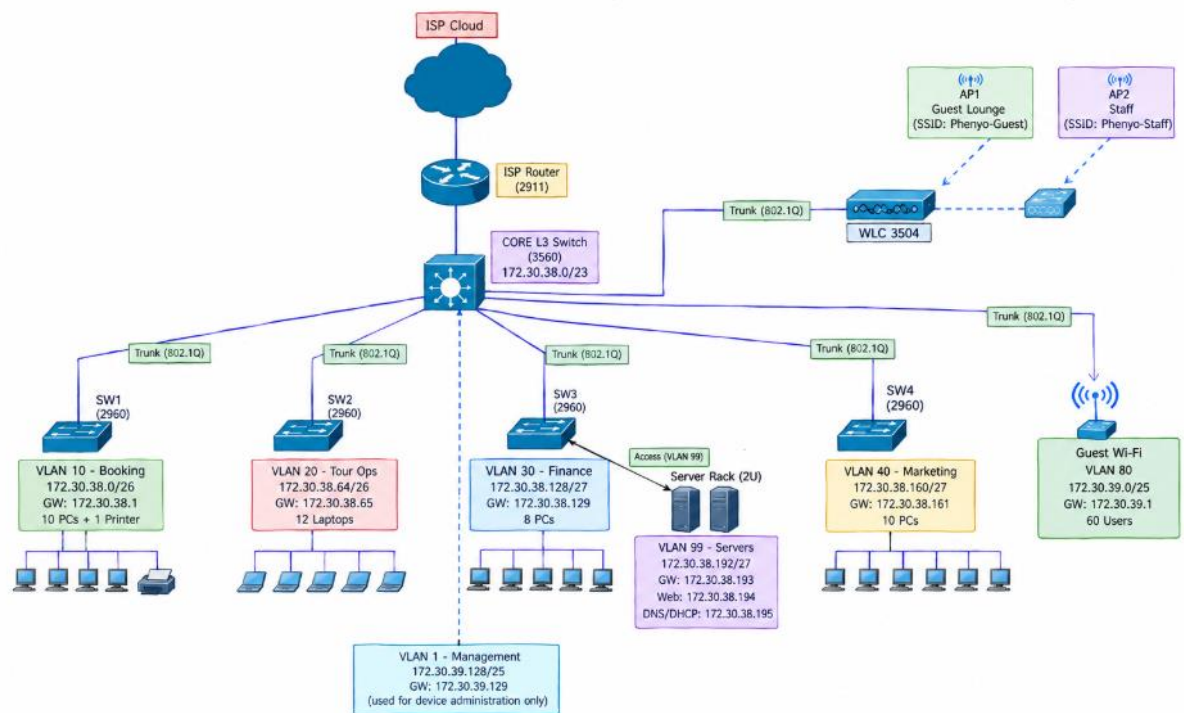
- Booking & Reservations desk (30 users) - needs access to Web booking system, fast internet, printer.
- Tour Operations / Guides (30 users) - mobile laptops, shared files.
- Finance & Admin (15 users) - highly confidential, isolated.
- Marketing (15 users) - media uploads, social media.
- Server Farm - Website [www.phenyotours.co.za](http://www.phenyotours.co.za), DNS, DHCP, Email.
- Guest Wi-Fi for customers in lounge (60 users) - isolated from staff network.

#### Technical Constraints from Brief:

- **Constraint:** All device administration must be secured - Previous incident of unauthorised changes. -> Solution: enable secret, service password-encryption, console/VTY passwords, SSH only, local admin account, banner.
- **Challenge:** Port Security (Foundational - switchport access control) -> Solution: Sticky MAC, max 2 MACs per access port, violation restrict.
- **CR5:** User numbers grow by 25% without renumbering -> Solution: VLSM over-provisioning with slack in each subnet + reserve subnet.

Addressing Block: 172.30.38.0/23 = 255.255.254.0 = 172.30.38.0 to 172.30.39.255 (510 usable)

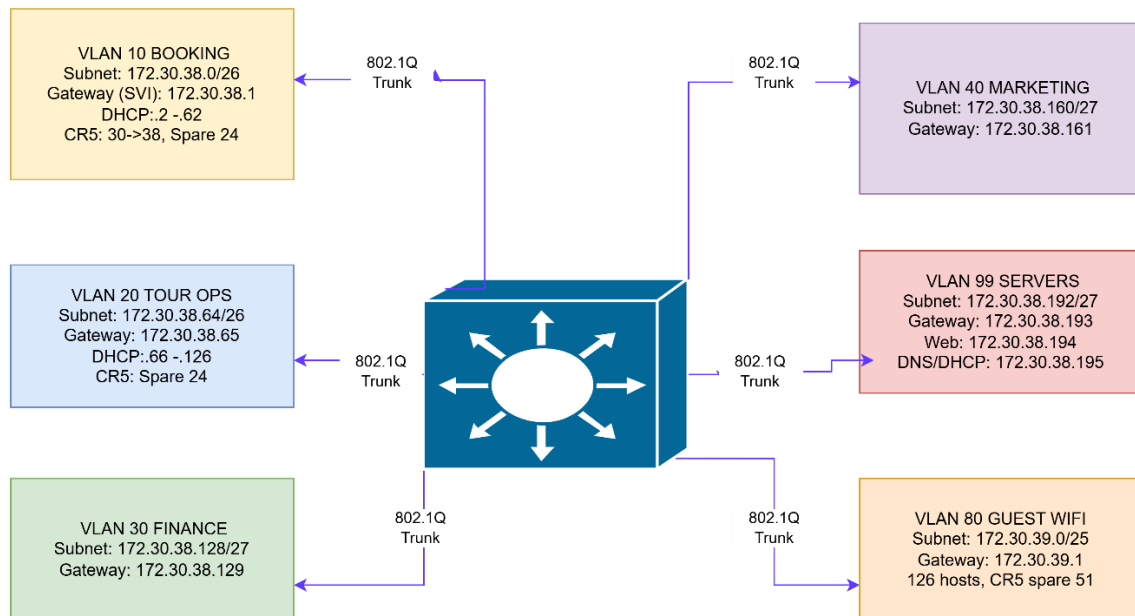
## 2. PHYSICAL TOPOLOGY



### Physical Layout:

- **Building:** Single HQ in Potchefstroom with 4 departments
- **Cabinet:** Core L3 Switch 3560 as central point, ISP Router 2911 for Internet
- **Access:** 4x Cisco 2960 Switches - One per department
  - SW1 Booking -> 10x desktops + printer
  - SW2 Tour Ops -> 12x laptops
  - SW3 Finance -> 8x desktops
  - SW4 Marketing -> 10x desktops
- **Server Rack:** 2U - Web Server, DNS/DHCP Server connected to Finance switch but in VLAN 99.
- **Wireless:** WLC 3504 + 2x AP - AP1 Guest Lounge (SSID Phenyo-Guest), AP2 Staff (SSID Phenyo-Staff)
- **Cabling:** All inter-switch links are GigabitEthernet Trunk (802.1Q), Access ports FastEthernet

## LOGICAL TOPOLOGY



| VLAN ID | VLAN Name       | Subnet Gateway   | IP(SVI)       | Subnet Mask Host Range | DHCP Pool                   |
|---------|-----------------|------------------|---------------|------------------------|-----------------------------|
| 10      | Booking         | 172.30.38.0/26   | 172.30.38.1   | 255.255.255.192 (/26)  | 172.30.38.2-172.30.38.62    |
| 20      | Tour Operations | 172.30.38.64/26  | 172.30.38.65  | 255.255.255.192 (/26)  | 172.30.38.70-172.30.38.126  |
| 30      | Finance         | 172.30.38.128/27 | 172.30.38.129 | 255.255.255.224 (/27)  | 172.30.38.130-172.30.38.158 |
| 40      | Marketing       | 172.30.38.160/27 | 172.30.38.161 | 255.255.255.224 (/27)  | 172.30.38.162-172.30.38.190 |
| 99      | Severs          | 172.30.38.192/27 | 172.30.38.193 | 255.255.255.224 (/27)  | 172.30.38.196-172.30.38.222 |

### VLAN Design:

- VLAN 10 - BOOKING - 172.30.38.0/26
- VLAN 20 - TOURS - 172.30.38.64/26

- VLAN 30 - FINANCE - 172.30.38.128/27
- VLAN 40 - MARKETING - 172.30.38.160/27
- VLAN 99 - SERVERS - 172.30.38.192/27
- VLAN 80 - GUEST WIFI - 172.30.39.0/25
- VLAN 1 - MGMT - 172.30.39.128/25 (unused for users)

**Routing:** Inter-VLAN routing on Core L3 using SVIs as default gateways. All Access switches use 802.1Q trunks to Core.

**Services:**

- Centralized DHCP on 172.30.38.195 serving all VLANs via ip helper-address
- DNS local: phenyotours.local
- Web: 172.30.38.194 -> [www.phenyotours.co.za](http://www.phenyotours.co.za)

**4. IP ADDRESSING PLAN - VLSM WITH CR5 PROOF**

| VLAN | Dept        | Current | +25%<br>=Required | Allocated<br>Subnet  | Mask            | Usable | Range         | Gateway            | Spare<br>CR5 |
|------|-------------|---------|-------------------|----------------------|-----------------|--------|---------------|--------------------|--------------|
| 10   | Booking     | 30      | 38                | 172.30.38.0/<br>26   | 255.255.255.192 | 62     | .2-<br>.62    | 172.30.38.<br>.1   | +24          |
| 20   | Tour<br>Ops | 30      | 38                | 172.30.38.64/<br>26  | .192            | 62     | .66-<br>.126  | 172.30.38.<br>.65  | +24          |
| 30   | Finance     | 15      | 19                | 172.30.38.128/<br>27 | .224            | 30     | .130-<br>.158 | 172.30.38.<br>.129 | +11          |
| 40   | Marketing   | 15      | 19                | 172.30.38.160/<br>27 | .224            | 30     | .162-<br>.190 | 172.30.38.<br>.161 | +11          |
| 99   | Servers     | 10      | 13                | 172.30.38.192/<br>27 | .224            | 30     | .194-<br>.222 | 172.30.38.<br>.193 | +17          |
| 80   | Guest       | 60      | 75                | 172.30.39.0/<br>25   | .128            | 126    | .2-<br>.126   | 172.30.39.<br>.1   | +51          |

|         |        |    |    |                      |      |     |                   |                   |               |
|---------|--------|----|----|----------------------|------|-----|-------------------|-------------------|---------------|
| 1       | Mgmt   | 10 | 13 | 172.30.39.1<br>28/25 | .128 | 126 | .130<br>-<br>.254 | 172.30.39<br>.129 | +11<br>3      |
| Reserve | Future |    |    | 172.30.38.2<br>24/27 | .224 | 30  |                   |                   | CR5<br>buffer |

### **CR5 Verification**

**CR5 Verification:** The assigned 172.30.38.0/23 address block contains 512 total addresses, of which 510 are usable host addresses. The VLSM design divides this address space into appropriately sized subnets for Booking, Tour Operations, Finance, Marketing, Servers, Guest Wi-Fi, Management and the future reserve network.

Each department subnet is provisioned to accommodate at least 25% growth without changing its existing network address or subnet. Booking and Tour Operations each have 62 usable addresses, Finance, Marketing and Servers each have 30 usable addresses, Guest Wi-Fi has 126 usable addresses, and Management has 126 usable addresses. The future reserve subnet provides 30 additional usable addresses for expansion.

Therefore, the addressing plan supports the CR5 requirement of 25% growth without renumbering the existing networks. The reserve subnet provides additional addressing capacity for future expansion.

<https://github.com/musanicholas100-arch/CMPG325-2026-063-Phenyo-Cultural-Tours/tree/main>